

Activity: nested hypothesis tests

Group members:

In this class activity, we will continue working with the `Kids198` data from the `Stat2Data` package, which examines the relationship between Age and Height for children between the ages of 8 and 18. We are also interested in how that relationship differs for boys and girls. The `Kids198` dataset includes the following variables on 198 children:

- **Height:** Height (in inches)
- **Age:** Age (in months)
- **Sex:** 0=male, 1 = female

Researchers want to know whether there is *any* relationship between Age and Height, after accounting for Sex. They believe that the relationship between Age and Height depends on Sex, so they fit the model

$$\text{Height} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Questions

To investigate their research question, the researchers fit the full and reduced models and calculate ANOVA tables for each model. Note that in the R output, the *SSModel* component is broken up between the different explanatory variables. The **Residuals** line in the ANOVA table is what corresponds to *SSE*.

```
library(Stat2Data)
data(Kids198)
m1 <- lm(Height ~ Age*Sex, data = Kids198)
m2 <- lm(Height ~ Sex, data = Kids198)
```

```
anova(m1)
```

Response: Height

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Age	1	5005.2	5005.2	575.820	< 2.2e-16	***
Sex	1	295.8	295.8	34.035	2.235e-08	***
Age:Sex	1	227.2	227.2	26.136	7.599e-07	***
Residuals	194	1686.3	8.7			

```
anova(m2)
```

Response: Height

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sex	1	480.7	480.67	13.99	0.0002412 ***
Residuals	196	6733.9	34.36		

1. Using the R output, calculate the F-statistic for the researchers' hypothesis test. What are the degrees of freedom for the corresponding F distribution?